

Grade 3 Ontario Curriculum Data

**Likely, Unlikely, Impossible — Grade 3
Predictions**

Unit Overview: Likely, Unlikely, Impossible — Grade 3 Predictions

Grade: Grade 3 | **Strand:** Data

Target Expectations: D2.1, D2.2

Learning Goal: Children will use the 5-word likelihood scale, predict outcomes, test predictions across two populations, and predict whether mean and mode stay the same.

Lesson	Lesson Title	Learning Goal
Day 1	The Five-Word Scale	I can use all 5 likelihood words on the scale.
Day 2	Predict and Decide	I can use likelihood words to make a prediction.
Day 3	Two Populations Test	I can test a prediction by comparing two populations.
Day 4	Same Mean? Same Mode?	I can predict if mean and mode stay the same across two populations.
Day 5	The Big Predict Capstone	I can predict using the scale, then test with mean and mode.

Lesson 1: The Five-Word Scale

Goal: FIVE likelihood words sit on a SCALE: impossible, unlikely, equally likely, likely, certain.

Expectation: D2.1

Learning Goal: "I can use all 5 likelihood words on the scale."

Materials: M1_scale_strip, M2_event_cards, M7_vocab_chart, M10_unit_poster

1. Minds On (12 mins): Lucky Lou's Five-Word Scale

Lucky Lou: Friends, in Grade 3 we use FIVE words for likelihood — impossible, unlikely, equally likely, likely, certain. They line up on a SCALE! Lucky opens: 'WHERE on the scale?'

Bring children to the carpet. Lucky Lou the Loon puppet in green. 'Detective friends — last year we used 3 words: impossible, possible, certain. In Grade 3 we add TWO more: unlikely and equally likely. That makes FIVE!'

2. Action (25 mins): Sort the 24 Event Cards

- Hand out 6 Event Cards per pair.
- Pairs explain ONE card to teacher.
- Stretch: pairs find ONE event for each of the 5 zones.
- Help with EQUALLY LIKELY zone — emphasize 50/50.
- Circulate to record diagnostic.
- STRATEGY COMPARE: Two pairs share their scale placements for the same event.

3. Consolidation (13 mins): Lucky Lou's Scale Brag

Discuss:

- Which 2 words are NEW in Grade 3?
- What's EQUALLY LIKELY?
- Order the 5 words.
- Name an UNLIKELY event.

Exit: Lucky Lou's refrain — class chants: 'Lucky Lou asks: WHERE on the scale? I say: ___!' (3x).

Lesson 2: Predict and Decide

Goal: We use likelihood words to PREDICT and make INFORMED DECISIONS.

Expectation: D2.1

Learning Goal: "I can use likelihood words to make a prediction."

Materials: M1_scale_strip, M3_coin_dice_set, M7_vocab_chart, M8_prediction_chart

1. Minds On (12 mins): Maybe Mae's Coin Flip Predict

Maybe Mae: I'll flip a coin 10 times. PREDICT — how many heads? Use the SCALE! Mae opens: 'PREDICT, then TEST, then TELL!'

Bring children to the carpet. Maybe Mae the Cat puppet in striped shirt. 'Detective friends — yesterday we placed events on the scale. Today we PREDICT.'

2. Action (25 mins): Predict 6 Events

- Hand out 6 prediction scenarios.
- For each, write likelihood word + decision.
- Stretch: revise prediction with new info.
- Help with decision links — likely → do, unlikely → don't.
- Circulate to record formative.
- STRATEGY COMPARE: After pairs predict 2-3 scenarios, pause for share-out.

3. Consolidation (13 mins): Maybe Mae's Predict Brag

Discuss:

- What's a PREDICTION?
- Name a LIKELY event for tomorrow.
- What decision follows 'rain is LIKELY'?
- When do we REVISE a prediction?

Exit: Mae's prediction refrain — 'Maybe Mae says — PREDICT, then TEST, then TELL!' (3x).

Lesson 3: Two Populations Test

Goal: We test predictions by collecting data from TWO populations and comparing.

Expectation: D2.2

Learning Goal: "I can test a prediction by comparing two populations."

Materials: M4_population_chart, M7_vocab_chart, M10_unit_poster

1. Minds On (12 mins): Lucky Lou's Two Classes

Lucky Lou: Imagine TWO classes. Class A loves hockey. Class B loves soccer. PREDICT — same favourite? TEST! Lucky+Mae open: 'Class A, Class B — same math, different DATA!' Yesterday Mae predicted; today we compare TWO classes' data.

Bring children to the carpet. Lucky Lou puppet in green. 'Detective friends — today we test predictions across TWO POPULATIONS.'

2. Action (25 mins): Compare Two Populations

- Hand out 4 two-population data sets.
- Compare data and decide if prediction was right.
- Stretch: explain WHY same or different.
- Help with comparison technique — find favourite of each.
- Circulate to record formative.
- STRATEGY INVENT: Pairs decide HOW they will compare — by favourite only, by top 2, or by all rows.

3. Consolidation (13 mins): Lucky Lou's Two-Class Brag

Discuss:

- What's a POPULATION?
- Why TWO?
- When are populations DIFFERENT?
- When are they the SAME?

Exit: Lucky+Mae two-population refrain — 'Class A, Class B — same math, different DATA!' (3x).

Lesson 4: Same Mean? Same Mode?

Goal: We can predict whether MEAN and MODE will be the SAME or DIFFERENT across populations.

Expectation: D2.2

Learning Goal: "I can predict if mean and mode stay the same across two populations."

Materials: M4_population_chart, M5_mean_mode_chart, M9_calculator_set, M7_vocab_chart

1. Minds On (12 mins): Maybe Mae's Mean and Mode Test

Maybe Mae: TWO classes. Same favourite snack. PREDICT — same MEAN? Same MODE? Mae opens: 'PREDICT, then TEST, then TELL!'

Bring children to the carpet. Maybe Mae puppet with calculator. 'Detective friends — yesterday we tested favourites across two populations. Today we test MEAN and MODE.'

2. Action (25 mins): Predict Mean and Mode Across Two Classes

- Hand out 4 paired data sets.
- Calculate mean and mode for both sets in each pair.
- Stretch: explain why same or different.
- Help with mean: $\text{total} \div \text{count}$.
- Circulate to record formative.
- STRATEGY COMPARE: Some pairs find mean by adding all then dividing; some pairs estimate.

3. Consolidation (13 mins): Maybe Mae's Mean+Mode Brag

Discuss:

- What's MEAN?
- When do they STAY the same?
- When do they DIFFER?
- What's a 'similar' population?

Exit: Mae's prediction-then-test refrain — 'Maybe Mae says — PREDICT, then TEST, then TELL!' (3x).

Lesson 5: The Big Predict Capstone

Goal: Predictors use the SCALE plus mean and mode to PREDICT, TEST, and DECIDE.

Expectation: D2.1, D2.2

Learning Goal: "I can predict using the scale, then test with mean and mode."

Materials: M1_scale_strip, M2_event_cards, M3_coin_dice_set, M4_population_chart

1. Minds On (12 mins): Lucky Lou's Big Predict

Lucky Lou: Today's BIG case — predict, test, decide. Use scale words AND mean/mode! Lucky+Mae open: 'SCALE, PREDICT, COMPARE, TEST, TELL!'

Bring children to the carpet. Lucky Lou and Maybe Mae puppets in capes. 'Friends — today is our CAPSTONE. We use ALL we know: 5 likelihood words, predictions, two-population tests, mean, and mode.'

2. Action (25 mins): The Big Predict

- Hand out templates.
- Stations 1-5: scale, predict, populations, mean+mode, conclude.
- Stretch: write a 2-sentence conclusion using ALL 5 scale words OR mean + mode.
- Help with prediction word choice as needed.
- Circulate to record summative evidence.
- STRATEGY COMPARE at action step 3: two children share DIFFERENT ways they decided their conclusion.

3. Consolidation (13 mins): Lucky Lou's Big Predict Brag

Discuss:

- What are the 5 likelihood words?
- What's a prediction?
- What's a population?
- Same mean?

Exit: Capstone refrain — Lucky + Mae call: 'SCALE, PREDICT, COMPARE, TEST, TELL — that's PROBABILITY!' (3x).

Name: _____

Date: _____

Lucky Lou's Five-Word Scale

Learning Goal: I can use all 5 likelihood words on the scale.

Instructions for the Student:

1. Match each event to one of the 5 likelihood words.
2. Place 6 events on the scale strip in the right zone.
3. Pick ONE event you placed under UNLIKELY.

Likelihood Scale



Name: _____

Date: _____

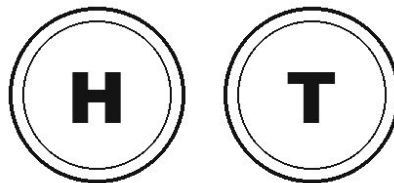
Maybe Mae's Predict and Decide

Learning Goal: I can use likelihood words to make a prediction.

Instructions for the Student:

1. Predict 4 coin/dice outcomes using a likelihood word.
2. For 4 weather/school scenarios, write a prediction word AND a decision.
3. ONE of your predictions changes.

2 Coins



H = Heads T = Tails

Name: _____

Date: _____

Lucky Lou's Two-Class Test

Learning Goal: I can test a prediction by comparing two populations.

Instructions for the Student:

1. For 4 scenarios, predict if Class A and Class B will have the SAME favourite.
2. Look at 2 paired data sets.
3. Compare your Part 1 prediction to the data.

Check Predictions

Check predictions vs data.

Name: _____

Date: _____

Maybe Mae's Mean and Mode

Learning Goal: I can predict if mean and mode stay the same across two populations.

Instructions for the Student:

1. For 4 paired data sets, predict if MEAN will be same/different and MODE will be same/different.
2. For 2 of the data sets, calculate the MEAN and MODE for each class.
3. Compare prediction to result.

Compare Results

Compare to actual results.

Name: _____

Date: _____

Lucky Lou's Big Predict

Learning Goal: I can predict using the scale, then test with mean and mode.

Instructions for the Student:

1. Read the scenario.
2. Look at Class A and Class B data.
3. Write a CONCLUSION using your scale word + the MEAN and MODE.

Predict Capstone

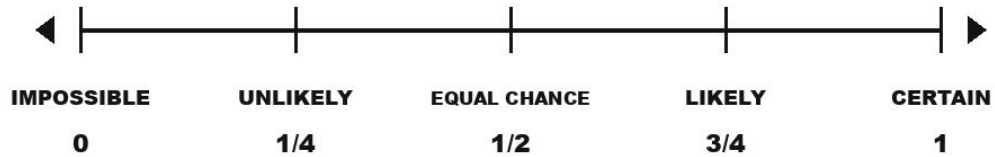
Capstone: predict outcomes.

Five-Word Scale Strip

(M1_scale_strip)

Letter-size strip with 5 likelihood zones (impossible, unlikely, equally likely, likely, certain).

Likelihood Scale



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 12 strips.
2. Laminate for dry-erase.
3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 strips
Lessons: 1, 2, 5
Prep: ~15 min

Likelihood Event Cards

(M2_event_cards)

24 event cards across 5 likelihood zones.

Event Cards

The sun rises tomorrow.	It will snow today.
I will pick a red marble.	I will roll a 7.
Coin lands on heads.	Tomorrow is Wednesday.
The ice will melt in the sun.	Dog will bark at noon.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 24 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 2, 5

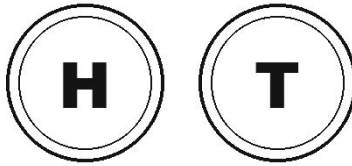
Prep: ~25 min

Coin and Dice Cards

(M3_coin_dice_set)

12 paper coins + 12 paper dice for prediction tests.

2 Coins



H = Heads T = Tails

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 24 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 2, 5

Prep: ~20 min

Two-Population Compare Chart

(M4_population_chart)

Letter-size chart with side-by-side Class A and Class B columns.

Outcome Table

Trial	Outcome	Tally	Probability

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 12 charts.

2. Laminate.

3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 charts

Lessons: 3, 4, 5

Prep: ~15 min

Mean and Mode Chart

(M5_mean_mode_chart)

Anchor chart of mean and mode formulas.

Mean / Mode
• MEAN: average of values. • MODE: most common value. • Use mean for typical value. • Use mode for most likely.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall before Lesson 4.

USE IN CLASS

Quantity: 1 chart

Lessons: 4, 5

Prep: ~10 min

Big Predict Template

(M6_capstone_template)

Letter-size capstone template with 5 sections.

Big Predict Template	
1	Scale _____
2	Predict _____
3	Populations _____
4	Mean+Mode _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

TEACHER PREP

1. Print 25 templates.

2. Stack near teacher table.

3. No lamination.

USE IN CLASS

Quantity: 25 templates

Lessons: 5

Prep: ~5 min

G3 Likelihood Vocabulary Chart

(M7_vocab_chart)

Anchor chart with G3 likelihood vocabulary.

G3 Likelihood Vocabulary Chart	
Impossible	Certain
● Unlikely ● Equally Likely ● Likely	● Prediction ● Population ● Mean

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall before Lesson 1.

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

Prediction-Decision Chart

(M8_prediction_chart)

Anchor chart linking likelihood to decisions.

Outcome Table

Trial	Outcome	Tally	Probability

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape next to vocab chart.

USE IN CLASS

Quantity: 1 chart

Lessons: 2, 5

Prep: ~10 min

Paper Calculator Cards

(M9_calculator_set)

12 paper calculator cards for mean+mode work.

Paper Calculator Cards	
1	Calculator With + _____
2	÷ Buttons _____
3	Each card _____
4	Calculator _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 12 cards per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

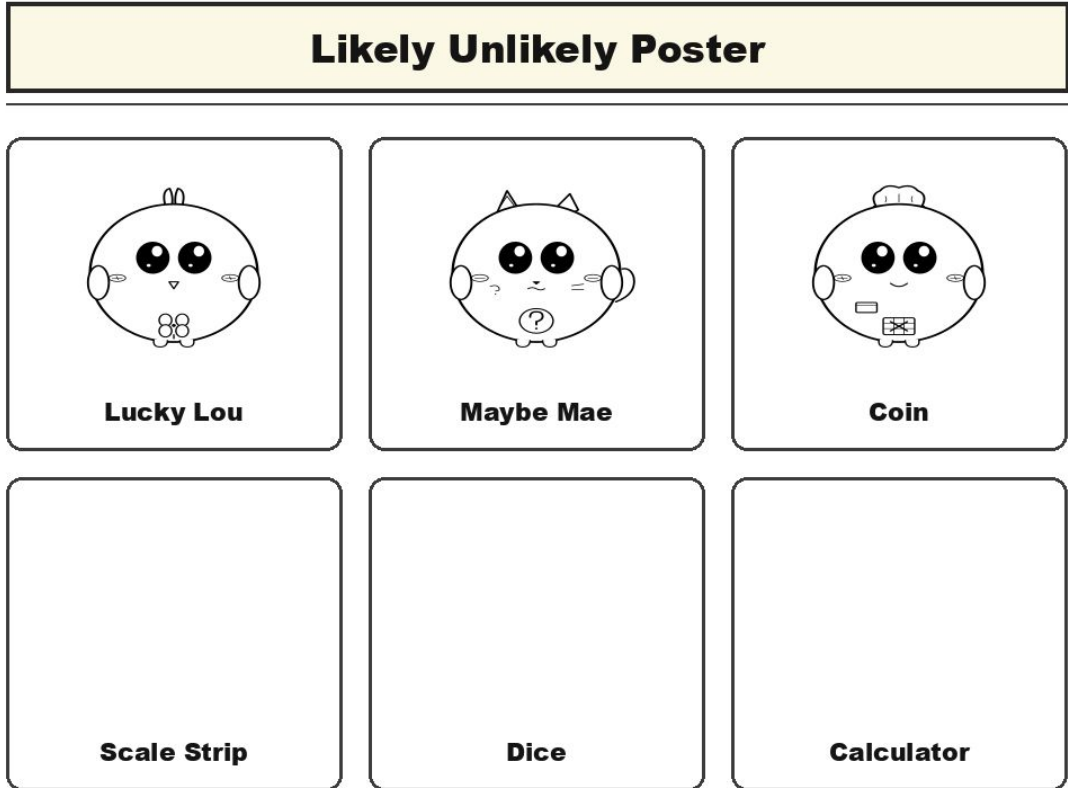
Lessons: 4, 5

Prep: ~15 min

Likely Unlikely Poster

(M10_unit_poster)

Unit anchor poster of Lucky Lou and Maybe Mae.



PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy on letter or 11x17.
2. Laminate.
3. Tape to front wall on Day 1.

USE IN CLASS

Quantity: 1 poster

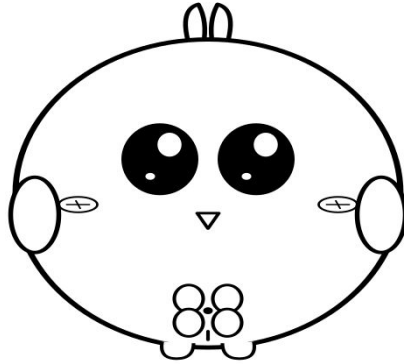
Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

Lucky Lou Puppet

(char_lucky_puppet)

Paper puppet of Lucky Lou the Loon for teacher use in all 5 lessons.



Lucky Lou the Loon

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette and tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

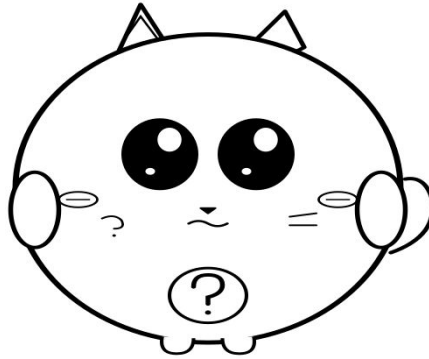
Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

Maybe Mae Puppet

(char_maybe_puppet)

Paper puppet of Maybe Mae the Cat for teacher use in Lessons 2-5.



Maybe Mae the Cat

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette and tab.

3. Laminate.

4. Glue popsicle stick.

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~12 min

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
D2.1 use mathematical language, including the terms 'impossible', 'unlikely', 'equally likely', 'likely', and 'certain', to describe the likelihood of events happening, and use that likelihood to make predictions and informed decisions	Uses yes/no without scale words. Cannot place events on scale.	Uses 2-3 of the 5 words with prompts. Some misplacements.	Independently uses all 5 likelihood words to describe events and make predictions.	Uses scale words to make predictions and explains the resulting decision.
D2.2 make and test predictions about the likelihood that the mean and the mode(s) of a data set will be the same for data collected from different populations	Compares one population only. No mean/mode prediction.	Predicts same/different with prompts. Computes one of mean or mode.	Independently predicts and tests mean and mode across two populations.	Explains why predictions were right or wrong using the data.

Junior Predictor Certificate (G3)

This certificate is proudly presented to:

For successfully completing the 5-Day G3 Likely Unlikely unit and demonstrating the skills of a G3 Junior Predictor.
Lucky Lou the Loon congratulates the student.

By completing this unit, this student has demonstrated:

- Using all 5 likelihood words on the scale
- Making predictions and decisions
- Comparing two populations
- Predicting and testing mean and mode

Marketplace Listing

A 5-day Grade 3 unit on probability framed around Lucky Lou and Maybe Mae.

Price: \$8.99 CAD

Pages: ~42

Classroom time: 250 min

Teacher prep: ~180 min

Top tags: Grade 3, math, probability, likelihood, prediction, mean, mode, two populations

What's included:

- Unit blueprint (1 file)
- 5 lesson plans (5 files)
- 5 worksheets (5 files)
- Manipulatives plan (1 file)
- Formative + reflection (1 file)
- Assessment suite (1 file)